

Understanding and Inspecting New Data Sets

Structured Data



What Makes Data Structured?

We've already learned that if data comes in a table format, it's **structured**.

But even within this format, some data sets are more structured than others.

Let's see some examples...

	A	B	C	D	E
1	Last Name	Sales	Country	Quarter	
2	Smith	\$16,753.00	UK	Qtr 3	
3	Johnson	\$14,808.00	USA	Qtr 4	
4	Williams	\$10,644.00	UK	Qtr 2	
5	Jones	\$1,390.00	USA	Qtr 3	
6	Brown	\$4,865.00	USA	Qtr 4	
7	Williams	\$12,438.00	UK	Qtr 1	
8	Johnson	\$9,339.00	UK	Qtr 2	
9	Smith	\$18,919.00	USA	Qtr 3	
10	Jones	\$9,213.00	USA	Qtr 4	
11	Jones	\$7,433.00	UK	Qtr 1	
12	Brown	\$3,255.00	USA	Qtr 2	
13	Williams	\$14,867.00	USA	Qtr 3	
14	Williams	\$19,302.00	UK	Qtr 4	
15	Smith	\$9,698.00	USA	Qtr 1	
16					



Take a look at the extract from a data set:

- What do you notice?
- What can we do to improve it?

Share your answers with the class.

Street Address
1234 Main Street, Sacramento, CA, 95822
5678 Broadway Avenue, Denver, CO, 80122
9810 Poplar Street, Philadelphia, PA 19108



How about now?

- What's changed?
- What makes this version better?

Street Address	City	State	Zip
1234 Main Street	Sacramento	CA	95822
5678 Broadway Avenue	Denver	CO	80122
9810 Poplar Street	Philadelphia	PA	19108

What Makes Data Structured?

With very structured data sets:

each **data point** or **observation** has its own row.

This makes the data easy to inspect, analyze and manipulate.

each **measure** or **variable** has its own column.



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1	Last Name	Sales	Country	Quarter	
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16					



Pair up with a classmate and take a look at the example below.

- What do you notice? What does this data set tell you?
- What can we do to improve it?

Country Name	Country Cod	Indicator Name	1960	1961	1962	1963
Aruba	ABW	Life expectancy at birth, total (years)	65.5693658	65.9880243	66.3655365	66.7139756
Andorra	AND	Life expectancy at birth, total (years)				
Afghanistan	AFG	Life expectancy at birth, total (years)	31.5800487	32.0959756	32.6118780	33.1273170
Angola	AGO	Life expectancy at birth, total (years)	32.9848292	33.3862195	33.7875853	34.1884634
Albania	ALB	Life expectancy at birth, total (years)	62.2543658	63.2734634	64.1628536	64.8870975
Arab World	ARB	Life expectancy at birth, total (years)	46.7626948	47.3886012	48.0024362	48.6075914
United Arab Emirates	ARE	Life expectancy at birth, total (years)	52.2432195	53.2865609	54.327	55.3635122
Argentina	ARG	Life expectancy at birth, total (years)	65.2155365	65.3385122	65.4326097	65.5093902
Armenia	ARM	Life expectancy at birth, total (years)	65.8634634	66.2843902	66.7098536	67.1378536

Data Formats | Columns and Rows

Here's a better way!

One row for each *variable*:

country name, country code,
year, and life expectancy.

It's okay if some data are
repeated!

Country Name	Country Code	Year	Life Expectancy
Aruba	ABW	1960	65.56936585
Aruba	ABW	1961	65.98802439
Aruba	ABW	1962	66.36553659
Aruba	ABW	1963	66.71397561
Afghanistan	AFG	1960	31.58004878
Afghanistan	AFG	1961	32.09597561
Afghanistan	AFG	1962	32.61187805
Afghanistan	AFG	1963	33.12731707
Angola	AGO	1960	32.98482927
Angola	AGO	1961	33.38621951
Angola	AGO	1962	33.78758537
Angola	AGO	1963	34.18846341